

REGINA H REYNOLDS

Molecular biologist turned bioinformatician, with experience translating large-scale omics data into actionable insights for drug discovery. I've led cross-functional analyses across transcriptomics, genomics, and functional screens, collaborating closely across teams to build reproducible pipelines that support impactful R&D decisions. I'm driven by a commitment to scientific rigour, integrity, and transparency – and by a broader desire to make a meaningful contribution to the world through thoughtful, data-driven science.

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CONTACT

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 [GitHub](#)

 [LinkedIn](#)

 [ResearchGate](#)



WORK EXPERIENCE

Present
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2025

Senior Principal Scientist, Bioinformatics Lead

AstronauTx

 London, UK

- Leading development of a modular EEG analytics platform to bring EEG analysis in-house in support of AstronauTx's sleep-based therapeutic approach. Built in Python and Nextflow, with each processing stage independently tested and validated against reference outputs.
- Contributing to translational understanding and biomarker strategy through QC, harmonisation, and analysis of historical and incoming EEG and other assay data.
- Establishing AstronauTx's computational foundations, spanning: cloud infrastructure; engineering practices; company-wide onboarding to Claude with guidance on output evaluation; and automation of manual processes (e.g. daily synchronisation of experimental timelines from the electronic lab notebook system into Monday.com via GitHub Actions).

2025
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2022

Lead Computational Biologist

CoSyne Therapeutics

 London, UK

- Provided scientific and strategic leadership as lead of a 5-person computational biology team, translating high-level objectives into clear milestones and actionable plans, managing tight deadlines and resources, and setting research priorities in collaboration with team members, experimental, engineering, and AI teams, and senior leadership.
- Drove company-wide 'omics initiatives spanning whole-genome sequencing, transcriptomics and CRISPRi screens to support target discovery in glioblastoma multiforme.
- Established scalable, robust, and reproducible computational workflows through adoption of tools such as Nextflow and Seqera Cloud for workflow orchestration, Docker for containerisation, R targets for reproducible exploratory data analysis, and Cruft templates for project scaffolding.
- Contributed to shaping company culture through line management; growing the computational biology team from 2 to 5 people; active involvement in recruitment across multiple teams; and contributing to the development of CoSyne's personal development framework. Additionally, hosted the nf-core community hackathon and spoke at Seqera Sessions, raising the team's external profile.

2025
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2022

Honorary Senior Research Fellow

University College London

 London, UK

- Acted as secondary supervisor to a PhD student and served on the thesis committee for another. Additionally contributed to analyses and provided input on papers across multiple research projects.

LANGUAGES

 R

 Bash

 Python

TOOLS

 Git/GitHub/GitLab

 Nextflow

 docker

*The source code is available
[GitHub](#).*

Last updated on 2026-08-17.

2022
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2021

Research Fellow

University College London

📍 London, UK

- Lead analyst involved in processing and analysing transcriptomic data to identify molecular signatures of Parkinson's disease progression. Worked within a large, multi-PI team and coordinated with researchers to align analytical approaches. Work performed using R, Nextflow, and Docker.
- Co-lead of Code and Pipeline Alignment Working Group in the [Aligning Sciences Across Parkinson's](#) initiative, facilitating the standardisation of data processing pipelines and coding practices across 10 international research teams, resulting in actionable recommendations to enhance data harmonisation and enable meta-analysis of post-mortem brain tissue datasets.
- Published 1 co-first author research article.

2016
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2016

Research Assistant

University of Copenhagen

📍 Copenhagen, Denmark

- Led project exploring the interactions between miR-34a, Sirt1 and p53 in a Huntington's disease mouse model, serving as lead experimentalist and analyst; work culminated in a first author [publication](#).
- In addition to these roles, facilitated collaboration with another research group and coordinated experiments with lab technicians, students, and researchers to support project goals.



EDUCATION

2021
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2016

PhD, Bioinformatics

University College London

📍 London, UK

- Thesis: [Exploring the importance of cell-type-specific gene expression regulation and splicing in Parkinson's disease](#).
- Integrated GWAS summary statistics with bulk/single-cell transcriptomic, eQTL, and chromatin accessibility data to identify cell-type-specific regulatory mechanisms in Parkinson's disease. Methods used included partitioned heritability, COLOC, and MAGMA.
- Applied transcriptomic methods, including cell-type deconvolution, differential splicing analysis, and RNA-binding protein motif analysis, to investigate splicing alterations and their relevance to Lewy body diseases.
- Published 3 first/co-first author research articles and 1 first author review. Successfully secured £10,000 from Signe og Peter Gregersens Mindefond to undertake transcriptional profiling of Parkinson's disease brain tissue.

2016
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2014

MSc, Molecular Biomedicine

University of Copenhagen

📍 Copenhagen, Denmark

- Thesis: Changes in the miR-34a-SIRT1 axis in Huntington's disease
- Grade: A (92.5%)

2013
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2010

BSc, Molecular Biomedicine

University of Copenhagen

📍 Copenhagen, Denmark

- Thesis: Pro-apoptotic factors in Huntington's disease: a study in the R6/2 transgenic mouse model
- Grade: A (96.7%)



TEACHING EXPERIENCE

2025
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2021

Subsidiary PhD Supervisor

University College London

📍 London, UK

- Secondary supervisor to Dr. Aaron Wagen. Provided support around transcriptomic analyses, bioinformatics workflows and statistical genetics.

2022
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2019

R fundamentals with [Clinician Coders](#)

University College London

📍 London, UK

- Developed [materials](#) and led workshops teaching basic R and tidy data principles to clinical academics.

2019
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2017

Omics Techniques

King's College London

📍 London, UK

- Lectured graduate level students on the principles of genome-wide association studies and led a workshop on how/why to use the Genotype-Tissue Expression portal.



VOLUNTARY WORK

Present
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2018

Peer Reviewer

📍 London, UK

- [Reviewer](#) for several scientific journals.

2022
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2017

Mentor

[Social Mobility Foundation](#)

📍 London, UK

- Mentored 4 A-level students looking to work in the field of biomedical research.



KEY PUBLICATIONS

2023

[Local genetic correlations exist among neurodegenerative and neuropsychiatric diseases](#)

NPJ Parkinson's disease

- Reynolds, RH, Wagen, AZ, Lona-Durazo, F, Scholz, SW, Shoai, M, Hardy, J, Gagliano Taliun, SA, Ryten, M
- Role: Co-first author, lead analyst and corresponding author.
- [Analysis](#) of local genetic correlations between neurodegenerative and neuropsychiatric disorders, with the aim of identifying genomic regions and genes that may drive pleiotropy.

2021

[Cross-platform transcriptional profiling identifies common and distinct molecular pathologies in Lewy body disorders](#)

Acta Neuropathologica

- Eleke, R, Reynolds, RH, Smith, A, Tilley, B, Gagliano Taliun, SA, Hardy, J, Matthews, PM, Gentleman, S, Owen, D, Johnson, MR, Srivastava, P, Ryten, M
- Role: Co-first author and analyst.
- [Transcriptomic analysis](#) of cell-type-specific changes in the Lewy body diseases.

A full list of publications is available online at <https://rhreynolds.github.io/cv>

- 2019 **Informing disease modelling with brain-relevant functional genomic annotations**
Brain
- Reynolds, RH, Hardy, J, Ryten, M, Gagliano Taliun, SA
 - Role: First author.
 - Review of conceptual advances in the generation of brain-relevant functional genomic annotations and among tools that allow integration of these annotations with genome-wide association summary statistics.
- 2019 **Moving beyond neurons: the Role of cell type-specific gene regulation in Parkinson's disease heritability**
NPJ Parkinson's disease
- Reynolds, RH, Botía, JA, Nalls, MA, International Parkinson's Disease Genomic Consortium (IPDGC), System Genomics of Parkinson's Disease (SGPD), Hardy, J, Gagliano Taliun, SA, Ryten, M
 - Role: First author and lead analyst.
 - Analysis of Parkinson's disease common variation, with the aim of identifying cell types and pathways of importance to disease risk.

CONFERENCES

- 2022 **AD/PD, Alzheimer's & Parkinson's Diseases Conference**
 Hybrid event
- Talk: Identifying genetic correlations among neurodegenerative and neuropsychiatric diseases
- 2021 **Genomics of Brain Disorders**
 Virtual event
- Talk: Dysregulation of splicing in human brain from individuals with Lewy body disease informs disease mechanisms
- 2019 **International Parkinson's Disease Genomics Consortium**
 London, UK
- Talk: Pairing bulk and single-nuclear RNA-seq to identify dementia-related pathways in PD
- 2019 **AD/PD, Alzheimer's & Parkinson's Diseases Conference**
 Lisbon, Portugal
- Talk: Mapping Parkinson's disease heritability to specific brain cell types
 - Received mention in a blog post on [Alzforum](#).
- 2018 **International Parkinson's Disease Genomics Consortium**
 Reykjavik, Iceland
- Talk: Moving beyond neurons: exploring the importance of cell type-specific gene expression in Parkinson's disease